

Bob Ross without ROS: A 3-Link Robotic Arm for Painting

Robotics Society Summer Project

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1 Project Overview

This project aims to develop a three-link planar robotic manipulator capable of reproducing drawings and paintings on a two-dimensional canvas. The system combines robotic kinematics, trajectory planning, image processing, and embedded control to convert digital images into executable robot motion.

Although the final demonstration focuses on robotic painting, the same concepts that are used to transform an image into smooth brush strokes are also used in industrial robot arms for manufacturing, autonomous inspection systems, surgical robots, and precision automation tasks. By building the complete pipeline from first principles, participants will gain an end-to-end understanding of how intelligent robotic systems perceive their environment, plan motion, and execute complex tasks in the physical world.

2 Problem Statement

Design and implement a three-link (3 DoF) robotic painting manipulator capable of converting image contours into executable trajectories and accurately reproducing them on a two-dimensional canvas.

The system must generate geometrically accurate paths while maintaining smooth joint motion and computational efficiency.

3 Project Goals and Approach

The primary goal of the project is to develop a complete image-to-painting robotic pipeline using custom kinematic and trajectory planning algorithms.

The project will begin with the development of a Python-based simulation environment. The trajectory generation strategy follows a hybrid planning approach. Desired paths will first be generated in task space to preserve geometric accuracy. These paths

will be represented as dense waypoint sets obtained from contour extraction and adaptive sampling.

For every sampled waypoint, a desired brush orientation will be generated from the local tangent of the contour. Inverse kinematics will then be solved only at these waypoints. This avoids repeated online inverse kinematics or Jacobian inversion during execution and shifts the computational burden to an offline planning stage.

After simulation validation, the algorithms will be deployed onto a physical three-link robotic arm. A host computer will perform image processing and trajectory generation, while an embedded microcontroller will receive and execute joint commands through serial communication.

The project also provides a foundation for future extensions such as obstacle avoidance, Jacobian-based path correction, energy-efficient trajectory optimization, and vision-based feedback control.

4 Tech Stack

Hardware

- Three-link planar robotic arm
- ESP32 microcontroller
- High-torque servo motors
- Pen or brush end-effector

Software

- Python
- NumPy
- SciPy
- OpenCV
- Matplotlib

5 Expectations from Contributors

Contributors are expected to dedicate 10-12 hours a week to:

- Attend weekly meetings
- Complete assigned tasks on time
- Understand theoretical frameworks used

6 Timeline and deliverables

Week	Milestone
1–2	PS Understanding + Forward kinematics implementation
3–4	Inverse kinematics implementation + Image processing for brush stroke extraction
5–6	Waypoint generation using brush strokes
7–8	Joint-space interpolation and trajectory execution in simulation
9–10	Hardware integration setup and calibration
10-12	End to end pipeline setup, robotic painting demonstration

Key Deliverables

- Interactive simulator capable of visualizing and controlling a 3-link robotic arm.
- Point-and-click target reaching using inverse kinematics and image contour extraction from input images.
- Image-to-waypoint pipeline capable of converting brush strokes into executable robot trajectories.
- Simulated robotic painter capable of reproducing drawings with smooth trajectory execution.
- End-to-end robotic painting system that converts an image into a physical painting on canvas.

7 References

1. OpenCV Documentation: <https://opencv.org>
2. NumPy Documentation: <https://numpy.org>
3. Peter Corke, “Robotics, Vision and Control.”

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